

Module/Pathway/Taxon Review: L-Histidine Biosynthesis in *Pseudomonas putida* KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Target bucket:** KEGG `ppu00340` "Histidine metabolism" → de novo L-histidine biosynthesis module (KEGG M00026, PRPP ⇒ histidine) **Module area:** amino_acid_metabolism **Scope of this review:** de novo L-histidine biosynthesis only (10 catalytic activities from PRPP + ATP to L-histidine)

1. Executive Summary

De novo L-histidine biosynthesis is **fully satisfiable** in *Pseudomonas putida* KT2440. All ten canonical catalytic activities that convert 5-phospho- α -D-ribose-1-diphosphate (PRPP) and ATP into L-histidine are encoded by identifiable candidate genes, with clean one-to-one KEGG Orthology (KO) and EC-number assignments for every biosynthetic step. The first committed step is present as a short-form ATP phosphoribosyltransferase (HisG, PP_0965) paired with a dedicated regulatory subunit (HisZ, PP_4890), the classic HisG/HisZ feedback-control architecture. The bifunctional imidazole-glycerol-phosphate synthase is split across HisF (PP_0293) and HisH (PP_0290); the amino-isomerase HisA (PP_0292), the dehydratase HisB (PP_0289), the aminotransferase HisC (PP_0967), and the terminal bifunctional dehydrogenase HisD (PP_0966) are all present; and the pyrophosphohydrolase/cyclohydrolase pair HisE (PP_5015) and HisI (PP_5014) covers the early steps.

The single genuine ambiguity is **step 8, histidinol-phosphate phosphatase (HolPase / HisN, EC 3.1.3.15)**. *P. putida* lacks the enteric bifunctional HisB (IGPD fused to a HAD-family phosphatase domain), so the phosphatase activity must be supplied by a stand-alone enzyme. Three paralogous loci carry HolPase-like annotations — PP_3157, PP_1721, and PP_5147 — and only one is the true HisN. Protein-family evidence resolves this cleanly: **PP_3157 belongs to the inositol-monophosphatase (IMPase) "his9" superfamily** that is the canonical non-enteric HolPase fold, carries the UniProt "Histidine biosynthesis" keyword and an explicit L-histidine-biosynthesis pathway assignment, and is corroborated by its GenBank product name

("Inositol monophosphatase family protein"). By contrast, PP_1721 and PP_5147 both belong to the HAD-like SerB family; PP_1721 is most likely a phosphoserine phosphatase (*serB*), and PP_5147 is a HAD hydrolase over-annotated with EC 3.1.3.15 but lacking the histidine-biosynthesis keyword. **PP_3157 should be curated as HisN; PP_1721 and PP_5147 should be excluded from the module.**

Finally, the parent KEGG bucket `ppu00340` conflates three distinct biological processes and **over-reaches** relative to the biosynthesis module. Seven candidate genes are catabolic or peripheral and should be excluded: the five histidine-utilization (*hut*) genes (*hutH*, *hutU*, *hutI*, *hutF*, *hutG*) that degrade histidine to glutamate, the glutathione-biosynthesis gene *gshA* (PP_0243), and PP_4983 (an amine oxidase mislabeled in local metadata as tryptophan 2-monooxygenase). All *his*-gene calls in this organism are **homology-inferred (UniProt PE=3)**; no in-strain biochemical or genetic validation of the histidine biosynthetic pathway exists for KT2440 itself.

2. Target-Organism Pathway Definition

What is included

The module scope is **de novo biosynthesis of L-histidine from PRPP and ATP** — the ten-activity linear route conserved across bacteria, archaea, fungi, and plants (KEGG module **M00026**, "Histidine biosynthesis, PRPP $\Rightarrow$ histidine"). In *P. putida* KT2440 this corresponds to the biosynthetic subset of KEGG map `ppu00340`. The pathway is metabolically distinctive because its purine-like intermediates connect it to nucleotide metabolism: the imidazole-glycerol-phosphate synthase step releases AICAR (5-aminoimidazole-4-carboxamide ribonucleotide), a purine-biosynthesis intermediate, recycling the adenine ring of ATP back into the nucleotide pool.

What must be kept separate

The KEGG map `ppu00340` "Histidine metabolism" is a broad overview map that mixes **three processes** that should not be co-curated into one biosynthesis module:

1. **De novo biosynthesis** (the *his* genes) — the target module.
2. **Histidine catabolism / utilization** (the *hut* genes): a five-step degradation route (histidase $\rightarrow$ urocanase $\rightarrow$ imidazolonepropionase $\rightarrow$ formimidoylglutamate deiminase $\rightarrow$ N-

formylglutamate deformylase) that converts histidine to glutamate for use as a carbon and nitrogen source. This is a catabolic locus, functionally opposite to biosynthesis.

3. **Peripheral / cross-listed reactions:** *gshA* (glutamate–cysteine ligase, glutathione biosynthesis) and PP_4983 (an amine oxidase). These appear on the overview map for chemical-neighborhood reasons, not because they participate in histidine biosynthesis.

Upstream **PRPP supply** (ribose-5-phosphate → PRPP, catalyzed by *prs* ribose-phosphate diphosphokinase) is shared central metabolism, not histidine-specific, and is correctly **outside** the candidate list and the module scope.

Alternate names / database definitions

- **KEGG:** map/module `ppu00340` (Histidine metabolism) and module `M00026` (Histidine biosynthesis, PRPP ⇒ histidine).
- **MetaCyc/BioCyc:** "L-histidine biosynthesis" (HISTSYN-PWY).
- **HisN nomenclature caveat:** the dedicated histidinol-phosphate phosphatase has been given at least three independent enzyme solutions across bacteria (enteric bifunctional HisB; PHP-family; IMPase/"his9" family). The gene symbol *hisN* was coined for the IMPase-type HolPase in *Corynebacterium glutamicum* ([PMID: 16901339](#)).

3. Expected Step Model

The canonical ten-activity model, mapped to *P. putida* KT2440 candidate genes:

Step	Reaction	Activity (EC)	Expected gene	KT2440 locus	KO	Call
0	R5P → PRPP	ribose-P diphosphokinase (2.7.6.1)	<i>prs</i>	(not in candidate list; shared metabolism)	K00948	out of scope
1	PRPP → PR-ATP	ATP phosphoribosyltransferase (2.4.2.17)	<i>hisG</i> (+ <i>hisZ</i>)	PP_0965 (+PP_4890)	K00765 (+K02502)	covered
2	PR-ATP → PR-AMP	phosphoribosyl-ATP pyrophosphohydrolase (3.6.1.31)	<i>hisE</i>	PP_5015	K01523	covered
3	PR-AMP → ProFAR	phosphoribosyl-AMP cyclohydrolase (3.5.4.19)	<i>hisI</i>	PP_5014	K01496	covered
4	ProFAR → PRFAR	ProFAR isomerase (5.3.1.16)	<i>hisA</i>	PP_0292	K01814	covered
5	PRFAR → IGP + AICAR	IGP synthase (4.3.2.10)	<i>hisF</i> + <i>hisH</i>	PP_0293 + PP_0290	K02500 + K02501	covered
6	IGP → IAP	IGP dehydratase (4.2.1.19)	<i>hisB</i>	PP_0289	K01693	covered
7	IAP → L-His-P	histidinol-P aminotransferase (2.6.1.9)	<i>hisC</i>	PP_0967	K00817	covered
8	L-His-P → L-histidinol	histidinol-P phosphatase (3.1.3.15)	<i>hisN</i>	PP_3157	K05602	covered (resolved)
9	L-histidinol → L-histidine	histidinol dehydrogenase (1.1.1.23)	<i>hisD</i>	PP_0966	K00013	covered

(Step numbering follows the biological order; UniProt's internal "step X/9" numbering differs by one because it collapses the PRPP-supply step.)

Nomenclature note (module_needs_revision, minor): the generic module outline inverts the HisE/HisI labels. Standard nomenclature is **HisE = phosphoribosyl-ATP pyrophosphohydrolase (EC 3.6.1.31)** acting first, and **HisI = phosphoribosyl-AMP cyclohydrolase (EC 3.5.4.19)** acting second. The provided outline labels step 3 "hisI diphosphatase" and step 4 "hisE cyclohydrolase," which is backwards. The *P. putida* loci themselves are correctly annotated (PP_5015 = HisE/EC 3.6.1.31; PP_5014 = HisI/EC 3.5.4.19); only the generic outline text needs the label swap.

```
PRPP —hisG(PP_0965)+hisZ(PP_4890)→ PR-ATP
      —hisE(PP_5015)→ PR-AMP —hisI(PP_5014)→ ProFAR
      —hisA(PP_0292)→ PRFAR —hisF(PP_0293)+hisH(PP_0290)→ IGP (+AICAR→purine pool)
      —hisB(PP_0289)→ IAP —hisC(PP_0967)→ L-His-P
      —hisN(PP_3157)→ L-histidinol —hisD(PP_0966)→ L-HISTIDINE
```

4. Candidate Genes and Evidence

High-confidence biosynthetic genes (covered)

hisG — PP_0965 (Q88P87), ATP phosphoribosyltransferase, EC 2.4.2.17, K00765. The first committed step. KEGG module M00026 encodes this step as the paired block **K00765–K02502**, i.e., a short-form catalytic HisG that requires the HisZ regulatory subunit. Evidence: homology (PE=3). Curation note: retain; expect HisZ-dependent activity and feedback control.

hisZ — PP_4890 (Q88DD7), ATP-PRT regulatory subunit, K02502. HisZ is a catalytically inactive paralog of histidyl-tRNA synthetase that partners with short-form HisG to form the active hetero-oligomeric ATP-PRT and confers L-histidine feedback inhibition on the complex. Its presence alongside a short-form HisG is internally consistent and the expected architecture. Evidence: homology. Curation note: retain as a required accessory subunit of step 1 (not an independent catalytic step).

hisE — PP_5015 (Q88D14), phosphoribosyl-ATP pyrophosphohydrolase, EC 3.6.1.31, K01523. Step 2. Evidence: homology (PE=3). Curation note: covered.

hisI — PP_5014 (Q88D15), **phosphoribosyl-AMP cyclohydrolase**, EC 3.5.4.19, K01496. Step 3. *hisI* and *hisE* are adjacent (PP_5014/PP_5015), consistent with the common HisIE gene neighborhood, though here annotated as separate monofunctional genes rather than a fusion. Evidence: homology (PE=3). Curation note: covered.

hisA — PP_0292 (Q88R42), **ProFAR isomerase**, EC 5.3.1.16, K01814. Step 4. Sits in the main biosynthetic cluster PP_0289–PP_0293. Evidence: homology. Curation note: covered.

hisF — PP_0293 (Q88R41), **IGP synthase cyclase subunit**, EC 4.3.2.10, K02500; and **hisH** — PP_0290 (Q88R44), **IGP synthase glutamine-amidotransferase subunit**, EC 4.3.2.10 (+3.5.1.2), K02501. Step 5, the two-subunit imidazole-glycerol-phosphate synthase. Both present in the PP_0289–PP_0293 cluster. Evidence: homology. Curation note: covered as an obligate heterodimer.

hisB — PP_0289 (Q88R45), **imidazoleglycerol-phosphate dehydratase**, EC 4.2.1.19, K01693. Step 6. Critically, in *P. putida* this is a **monofunctional IGPD** (dehydratase domain only), NOT the enteric bifunctional HisB that also carries a HAD-family HolPase domain. This is precisely why *P. putida* needs a separate HisN for step 8. Evidence: homology. Curation note: covered; flag as monofunctional.

hisC — PP_0967 (Q88P86), **histidinol-phosphate aminotransferase**, EC 2.6.1.9, K00817. Step 7. Local metadata lists its primary bucket as `ppu00401`, but this is only a secondary cross-listing of the promiscuous aminotransferase on another map; the UniProt "Histidine biosynthesis" keyword and internal pathway step (7/9) confirm this is the genuine histidine HisC. Curation note: covered; retain despite the ppu00401 bucket label.

hisD — PP_0966 (P59400), **histidinol dehydrogenase**, EC 1.1.1.23, K00013. Terminal step, the bifunctional NAD⁺-dependent dehydrogenase that performs two successive oxidations (histidinol → histidinal → histidine). This is the only candidate with a Swiss-Prot (reviewed) accession (P59400), though still PE=3. Adjacent to hisG (PP_0965) and hisC (PP_0967). Curation note: covered.

The resolved ambiguous step — HisN (step 8)

hisN — PP_3157 (Q88I44), **histidinol-phosphate phosphatase**, EC 3.1.3.15, K05602 — **HIGH CONFIDENCE**. *P. putida* lacks a bifunctional HisB HolPase domain, so a stand-alone HolPase is required. Three paralogs are annotated with EC 3.1.3.15 (see §5), but PP_3157 is the strongest candidate on protein-family grounds: it belongs to the **inositol-monophosphatase (IMPase) superfamily** — InterPro IPR000760 (IMPase-like), IPR011809 ("Histidinol-phosphatase His_9, proposed"), Pfam PF00459 (Inositol_P) — carries UniProt keywords "Histidine biosynthesis" /

"Amino-acid biosynthesis" and an explicit L-histidine-biosynthesis pathway/step assignment, and its GenBank product name is "Inositol monophosphatase family protein." The IMPase "his9" fold is the canonical non-enteric HolPase family established in *Corynebacterium glutamicum* and Actinobacteria (PMID: 16901339; PMID: 28720084). Curation note: **assign as HisN; promote to full fetch-gene review.**

Out-of-module candidates (not_expected_in_target_taxon for a biosynthesis module)

Gene	Locus	Annotation	KO/EC	Why excluded
hutH	PP_5032	histidine ammonia-lyase (histidase)	K01745 / 4.3.1.3	catabolism (His degradation)
hutU	PP_5033	urocanate hydratase (urocanase)	K01712 / 4.2.1.49	catabolism
hutI	PP_5030	imidazolonepropionase	K01468 / 3.5.2.7	catabolism
hutF	PP_5036	formimidoylglutamate deiminase	K05603 / 3.5.3.13	catabolism
hutG	PP_5029	N-formylglutamate deformylase	K01458 / 3.5.1.68	catabolism
gshA	PP_0243	glutamate–cysteine ligase	K01919 / 6.3.2.2	glutathione biosynthesis (peripheral)
PP_4983	PP_4983	amine oxidase (locally mislabeled Trp 2-monooxygenase)	K00274 / 1.4.3.4	peripheral; local EC 1.13.12.3 likely wrong

The *hut* genes cluster at PP_5029–PP_5036 and constitute the catabolic histidine-utilization locus, functionally the reverse of biosynthesis (PMID: 17717196).

5. Gaps, Ambiguities, and Likely Over-Annotations

The HolPase paralog problem (the central curation decision)

Three loci carry EC 3.1.3.15 (or related phosphatase) annotations. Only one is the true HisN. This is exactly the over-propagation hazard flagged by Kulis-Horn et al.: IMPase-like paralogs (CysQ, ImpA, SuhB) are highly similar to HisN yet do not perform HolPase in vivo ([PMID: 28720084](#)).

Locus	Family (InterPro/ Pfam)	GenBank product	UniProt "His biosynthesis" kw	Genomic neighborhood	Verdict
PP_3157 (Q88I44)	IMPase superfamily; IPR000760 / IPR011809 (His_9) / PF00459	"Inositol monophosphatase family protein"	Yes + pathway assignment	isolated (nbrs: universal-stress PP_3156, exported protein PP_3158)	true HisN
PP_5147 (Q88CN3)	HAD-like SerB; IPR006385 / IPR050582 / PF12710	"Hydrolase, haloacid dehalogenase-like family"	No	nbrs RNA pyrophosphohydrolase PP_5146, membrane protein PP_5148	HAD over-annotation
PP_1721 (Q88M53)	HAD-like SerB; IPR006385 / IPR050582 / PF12710	"putative phosphoserine phosphatase"	No (PE=4 Predicted; blank protein name)	nbrs Zn-ADH PP_1720, ABC transporter PP_1722	likely <i>serB</i> , not HolPase

The two decisive discriminators are (1) **protein fold** — PP_3157 is IMPase/his9 (the bona fide non-enteric HolPase fold), whereas PP_5147 and PP_1721 are HAD-SerB; and (2) **the UniProt "Histidine biosynthesis" keyword plus explicit pathway assignment**, which only PP_3157 carries. Dispersal of PP_3157 away from the *his* clusters is expected and normal for IMPase-type HisN, which is typically not co-located with the biosynthetic operon. Two independent annotation sources (UniProt InterPro families and KEGG/GenBank product names) agree.

Curation actions: PP_5147 → mark `candidate_uncertain` / likely over-annotation (HAD hydrolase; strip the histidine-biosynthesis inference). PP_1721 → re-annotate as *serB* (phosphoserine phosphatase); remove from the histidine module.

Local-metadata errors flagged for correction

- **PP_4983** is labeled "Tryptophan 2-monooxygenase (EC 1.13.12.3)" in local metadata, but KEGG assigns K00274 (monoamine oxidase, EC 1.4.3.4). Peripheral regardless; reconcile the EC to the KEGG call and exclude from the histidine module.
- **PP_1721** local annotation "Phosphoserine phosphatase (EC 3.1.3.-)" is consistent with the SerB re-assignment — the local EC 3.1.3.- (not 3.1.3.15) already hints it is not a dedicated HolPase.
- **Generic outline HisE/HisI label inversion** (see §3): a documentation fix, not a gene-level error.

Evidence-level gap

Every *his* candidate is **UniProt PE=3 (inferred from homology)**. There is no direct biochemical or genetic characterization of the histidine biosynthetic pathway in KT2440 itself. All calls rest on sequence homology, KO/EC transfer, and protein-family membership.

6. Module and GO-Curation Recommendations

Per-step module status

Module step	Status	Gene(s)
1 ATP-PRT (HisG + HisZ)	covered	PP_0965 (+ PP_4890)
2 PR-ATP pyrophosphohydrolase (HisE)	covered	PP_5015
3 PR-AMP cyclohydrolase (HisI)	covered	PP_5014
4 ProFAR isomerase (HisA)	covered	PP_0292
5 IGP synthase (HisF + HisH)	covered	PP_0293 + PP_0290
6 IGP dehydratase (HisB, monofunctional)	covered	PP_0289
7 HolP aminotransferase (HisC)	covered	PP_0967
8 HolPase (HisN)	covered (resolved by fold evidence)	PP_3157
9 histidinol dehydrogenase (HisD)	covered	PP_0966
— HolPase paralogs	candidate_uncertain / exclude	PP_5147, PP_1721
— <i>hut</i> catabolism (×5), <i>gshA</i> , PP_4983	not_expected_in_target_taxon	PP_5029/30/32/33/36, PP_0243, PP_4983

Overall module verdict: fully covered, 10/10 catalytic steps satisfied.

Module boundary and document recommendations

1. **Split the bucket.** The generic module boundary that inherits all of `ppu00340` is wrong for a biosynthesis module. Recommend a dedicated biosynthesis module (M00026-equivalent) containing only PP_0965, PP_4890, PP_5015, PP_5014, PP_0292, PP_0293, PP_0290, PP_0289,

PP_0967, PP_3157, PP_0966. The *hut* catabolic locus warrants its own separate catabolism module.

2. **Fix the HisE/HisI outline labels** (module_needs_revision, minor/documentation).
3. **GO curation for PP_3157:** support annotation to GO:0004401 (histidinol-phosphatase activity) and GO:0000105 (histidine biosynthetic process). Because the assignment is homology-based, use evidence code **IEA/ISS**, not experimental codes.
4. **GO down-annotation for PP_5147 and PP_1721:** remove or qualify GO:0004401; PP_1721 should carry GO:0004647 (phosphoserine phosphatase activity) / serine biosynthesis.
5. **No new GO terms appear necessary** — the required terms (histidinol-phosphatase activity; histidine biosynthetic process; ATP-PRT regulatory activity for HisZ) already exist.

7. Genes to Promote to Full fetch-gene Review

1. **PP_3157 (HisN) — highest priority.** The one step resolved by inference rather than a named dedicated gene. A full review should confirm the IMPase/his9 active-site residues, verify it is not a CysQ/ImpA/SuhB-type IMPase with a different substrate, and (ideally) seek experimental or structural evidence. This is the linchpin of module satisfiability.
2. **PP_1721 and PP_5147 — paired disambiguation review.** Confirm PP_1721 = *serB* and PP_5147 = generic HAD hydrolase, and formally strip their HolPase over-annotations.
3. **PP_4983 — EC reconciliation.** Resolve the tryptophan-2-monooxygenase (local) vs. monoamine-oxidase (KEGG) discrepancy; confirm exclusion from the histidine module.
4. **PP_0965 / PP_4890 (HisG/HisZ pair).** Lower priority; confirm the short-form HisG requires HisZ and document the feedback-regulation architecture.

The remaining core *his* genes (PP_5015, PP_5014, PP_0292, PP_0293, PP_0290, PP_0289, PP_0967, PP_0966) have clean, unambiguous KO/EC/keyword assignments and do not need individual promotion unless the module is being upgraded from IEA to experimental evidence.

8. Evidence Base

PMID	Title (abbrev.)	Organism	How it supports the review
16901339	<i>IS6100 transposon mutagenesis identifies the last His gene in C. glutamicum</i>	<i>C. glutamicum</i>	Established that the IMPase-like gene (cg0910, renamed hisN) is the missing HolPase (EC 3.1.3.15) in a non-enteric bacterium — the precedent for assigning an IMPase-fold gene (PP_3157) as HisN. Direct experimental evidence; species transfer moderate (same fold family, different genus).
28720084	<i>Sequence-based identification of IMPase-like HisN in C. glutamicum, Actinobacteria, and beyond</i>	Actinobacteria + survey	Defines the IMPase/"his9" HolPase family and explicitly warns that IMPase paralogs (CysQ, ImpA, SuhB) are very similar to HisN but most likely do not participate in histidine biosynthesis — the exact over-propagation hazard motivating disambiguation of PP_3157 vs. PP_5147/PP_1721.
17717196	<i>Genetic analysis of the hut genes in P. fluorescens SBW25</i>	<i>P. fluorescens</i>	Shows the hut locus is catabolic (histidine as sole C/N source), supporting exclusion of hutH/U/I/F/G from the biosynthesis module. Species transfer to <i>P. putida</i> : strong (same genus, orthologous locus).
9561727	<i>hut operon upregulated at low temperature in P. syringae</i>	<i>P. syringae</i> (+ <i>P. putida</i> assays)	Confirms <i>hut</i> genes (hutU urocanase, histidase) are a regulated catabolic operon in pseudomonads, including direct enzyme assays in <i>P. putida</i> — reinforces the biosynthesis/catabolism split.
2203754	<i>hutC repressor sequence, K. aerogenes (with P. putida comparison)</i>	<i>K. aerogenes</i> / <i>P. putida</i>	Documents <i>hut</i> operon regulation and the <i>P. putida</i> <i>hutC</i> repressor, corroborating that the <i>hut</i> genes form a distinct regulated catabolic system.

Direct vs. inferred evidence summary: No cited study characterizes the *P. putida* KT2440 histidine biosynthetic genes directly. The strongest experimental anchor

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P 16901339

) is the *C. glutamicum* HisN identification, transferred by fold homology. The *hut*/catabolism exclusions are supported by direct genus-level experiments (PMIDs 17717196, 9561727, 2203754). All histidine biosynthetic assignments in KT2440 remain homology-inferred.

9. Limitations and Knowledge Gaps

- **No in-strain experimental validation.** All *his* candidates are UniProt PE=3 (inferred from homology). There is no published biochemical assay, knockout auxotrophy, or structure for any KT2440 histidine biosynthetic enzyme. Module satisfiability is therefore an inference, albeit a strong one.
- **HisN rests on fold + keyword inference, not function.** PP_3157 is the best candidate, but the assignment relies on IMPase/his9 family membership and UniProt keywords rather than a demonstrated HolPase activity in *P. putida*. A CysQ/ImpA/SuhB-type false positive cannot be fully excluded without functional data (the caution raised by

P 28720084

- **Regulatory architecture unverified.** The HisG(short)/HisZ feedback model is inferred from the KO pairing (K00765–K02502); *P. putida* HisZ-dependence and histidine feedback inhibition have not been demonstrated in-strain.
- **Local-metadata inconsistencies** (PP_4983 EC label; PP_1721 annotation; generic HisE/HisI outline inversion) indicate the source metadata is imperfect and should not be treated as ground truth.
- **PRPP flux/regulation** (upstream *prs*) and the AICAR-recycling link to purine metabolism were not analyzed quantitatively; they are out of module scope but relevant to any flux-based satisfiability claim.

10. Proposed Follow-up Experiments / Actions

1. **Functional confirmation of PP_3157 as HisN.** Construct a PP_3157 deletion in KT2440 and test for histidine auxotrophy rescuable by histidinol (the assay design that identified *C. glutamicum hisN*, P 16901339). Complement in a HolPase-deficient *E. coli*/*C. glutamicum* background. This is the single most valuable experiment for the module.
2. **In vitro HolPase assay** on purified PP_3157 (with PP_5147 and PP_1721 as negative controls) against L-histidinol-phosphate to directly measure EC 3.1.3.15 activity and rule out IMPase off-target activity.
3. **serB confirmation for PP_1721.** Test complementation of an *E. coli serB* mutant and/or serine auxotrophy to formally re-annotate PP_1721 and retire its HolPase label.
4. **Curation updates now (no experiment needed):** (a) assign PP_3157 = HisN with GO:0004401/GO:0000105 (IEA/ISS); (b) exclude PP_5147, PP_1721 from the histidine module and re-annotate PP_1721 as serB; (c) split the biosynthesis module from the *hut* catabolism genes and *gshA*/PP_4983; (d) fix the generic-outline HisE/HisI label inversion; (e) reconcile the PP_4983 EC number to KEGG K00274.
5. **HisG/HisZ regulation.** Enzyme assay of the reconstituted PP_0965/PP_4890 complex $\pm$ L-histidine to confirm HisZ-dependent activity and feedback inhibition.

Bottom line

De novo L-histidine biosynthesis is **fully satisfiable (10/10 steps)** in *P. putida* KT2440. The only non-trivial call — the stand-alone histidinol-phosphate phosphatase — resolves to **PP_3157 (IMPase/his9-family HisN)**, with PP_5147 and PP_1721 excluded as HAD/SerB over-annotations. The parent ppu00340 bucket over-reaches and should be narrowed to exclude the *hut* catabolic genes, *gshA*, and PP_4983. All assignments are homology-based and would benefit from targeted experimental confirmation, most urgently a PP_3157 auxotrophy/complementation test.